

Simulator-Verified Reasoning Distillation for Causal HAPS–ISAC Control

Mathematical Formulation and Implementation Specification

Companion optimisation document

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Abstract

This document specifies an implementable optimisation framework for the previously formulated freshness-aware secure HAPS–ISAC system. A large reasoning model is used only as an offline planning teacher. Given a compact representation of a network state, the teacher proposes several high-level mixed discrete–continuous actions. A physics-based action-completion module constructs communication, sensing, and jamming beamformers; a feasibility layer enforces hard constraints; and stochastic simulator rollouts rank the candidates. The resulting verified demonstrations are distilled into a lightweight numerical student policy. Dataset aggregation on student-visited states and constrained proximal policy optimisation then correct distribution shift and permit improvement beyond an imperfect teacher.

1 Purpose and design principles

The framework solves the long-term causal problem

$$\min_{\pi} J(\pi) = \omega_I \widetilde{\Delta}^I + \omega_S \widetilde{\Delta}^S + \omega_U \widetilde{U} + \omega_E \widetilde{E}, \quad (1)$$

subject to scheduling, NOMA, power, sensing, RIS, secrecy, AoLI, reliability, computation, mobility, energy, and causal-information constraints. The design uses five principles:

1. the teacher proposes decisions but never certifies its own optimality;
2. every proposal is completed, repaired, and evaluated by the simulator;
3. the model controls a compact high-level action, not complex beam entries;
4. long-term constraints are exposed through virtual queues or dual variables;
5. deployment uses a small numerical policy with a deterministic safety layer.

$$\boxed{\text{state} \rightarrow \text{teacher} \rightarrow \text{candidates} \rightarrow \text{physics completion} \rightarrow \text{repair} \rightarrow \text{rollout verification} \rightarrow \text{student}.} \quad (2)$$

2 Constrained belief-state representation

2.1 Causal state

At the beginning of slot t , define

$$\mathbf{s}_t = [\mathbf{o}_1[t], \dots, \mathbf{o}_M[t], \mathbf{o}_g[t], \mathbf{z}[t], \mathbf{c}[t]]. \quad (3)$$

$\mathbf{o}_m$ is a pair token, $\mathbf{o}_g$ is the global physical state, $\mathbf{z}$ contains virtual queues, and $\mathbf{c}$ is an optional objective context. Only information available before the action is permitted; the echo generated during slot t cannot appear in $\mathbf{s}_t$.

2.2 Pair and global tokens

For pair m , use the 14-dimensional token

$$\mathbf{o}_m = [\widetilde{\Delta}_{m,n}^I, \widetilde{\Delta}_{m,f}^I, \widetilde{\Delta}_{m,n}^L, \widetilde{\Delta}_{m,f}^L, \widetilde{G}_{H,m,n}, \widetilde{G}_{H,m,f}, \widetilde{C}_{m,n}, \widetilde{C}_{m,f}, \widetilde{G}_{J,m,n}, \widetilde{G}_{J,m,f}, \widetilde{L}_{m,n}, \widetilde{L}_{m,f}, \widetilde{\mu}_m^{\text{SIC}}, \widetilde{\tau}_m^{\text{wait}}]^\top \in \mathbb{R}^{14}. \quad (4)$$

$G_{H,m,k} = \|\widehat{\mathbf{g}}_{m,k}\|^2$ is a HAPS–UE strength, $C_{m,k} = \text{tr}(\mathbf{C}_{m,k})$ is CSI uncertainty, $G_{J,m,k} = \|\widehat{\mathbf{f}}_{J,m,k}\|^2$ is an AAV interference proxy, μ_m^{SIC} is a pre-beamforming SIC-margin proxy, and τ_m^{wait} is time since service.

Use the 24-dimensional global vector

$$\mathbf{o}_g = [\widetilde{\Delta}^S, \widetilde{\mathbf{x}}_T^{\text{av}} \in \mathbb{R}^4, \text{vech}(\widetilde{\mathbf{P}}^{\text{av}}) \in \mathbb{R}^{10}, \widetilde{\mathbf{q}}_J - \widetilde{\mathbf{q}}_T \in \mathbb{R}^3, \widetilde{\mathbf{v}}_J \in \mathbb{R}^2, \widetilde{E}_J, \widetilde{\rho}_{\text{SI}}, b_R, b_J]^\top \in \mathbb{R}^{24}. \quad (5)$$

b_R and b_J indicate whether RIS and AAV mobility actions may change. The previous RIS codeword is supplied through a learned embedding. Features are percentile-clipped and scaled to $[-1, 1]$ or $[0, 1]$.

2.3 Objective context

For a student that must follow changing intents, include

$$\mathbf{c} = [\omega_I, \omega_S, \omega_U, \omega_E, \{\tilde{\Delta}_{m,k}^{L,\min}\}_{m,k}, \{\tilde{R}_{m,k}^{\text{sec},\min}\}_{m,k}, \tilde{P}^{\max}, \tilde{E}_{\text{reserve}}]. \quad (6)$$

For a fixed experiment, omit $\mathbf{c}$ and keep these values in configuration.

3 Long-term constraints through virtual queues

For each average constraint $g_j(\mathbf{s}_t, \mathbf{a}_t) \leq 0$, define

$$Z_j[t+1] = [Z_j[t] + g_j(\mathbf{s}_t, \mathbf{a}_t)]^+, \quad Z_j[0] = 0. \quad (7)$$

Examples are

$$g_{m,k}^L[t] = \Delta_{m,k}^{L,\min} - \Delta_{m,k}^L[t], \quad (8)$$

$$g_{m,k}^\chi[t] = \chi_{m,k}^{\min} - \chi_{m,k}[t], \quad (9)$$

$$g_{m,k}^{\text{sec}}[t] = a_m[t](\mathbf{1}\{R_{m,k}^{\text{sec}}[t] < R_{m,k}^{\text{sec},\min}\} - \epsilon_{m,k}^{\text{sec}}), \quad (10)$$

$$g^{\text{sen}}[t] = \mathbf{1}\{\gamma_s[t] < \Gamma_s\} - \epsilon_s, \quad (11)$$

$$g^P[t] = \text{tr}(\mathbf{P}^{\text{av}}[t]) - P^{\max}. \quad (12)$$

The normalised queue vector is

$$\mathbf{z}[t] = \text{clip}([Z_1/Z_1^{\text{ref}}, \dots, Z_{N_g}/Z_{N_g}^{\text{ref}}], 0, z_{\max}). \quad (13)$$

Equivalently, dual variables may be updated as

$$\lambda_j[t+1] = [\lambda_j[t] + \eta_\lambda g_j[t]]^+. \quad (14)$$

Do not use queues and duals simultaneously unless their roles are separated.

4 High-level mixed action

The teacher and student predict

$$\mathbf{a}_t^H = (m_t, c_{R,t}, \eta_{H,t}, \eta_{c,t}, \eta_{n,t}, \eta_{J,t}, \psi_{J,t}, \nu_{J,t}, \eta_{f,t}), \quad (15)$$

where

$$\begin{aligned} m_t &\in \{0, \dots, M\}, & c_{R,t} &\in \{0, \dots, C_R\}, \\ \eta_H, \eta_c, \eta_J, \eta_f &\in [0, 1], & \eta_n &\in [0, 0.5], \\ \psi_J &\in [-\pi, \pi], & \nu_J &\in [0, V_J^{\max}]. \end{aligned} \quad (16)$$

$m_t = 0$ means no payload and $c_{R,t} = 0$ means retain the RIS. Map

$$P_H^{\text{use}} = \eta_H P_H^{\max}, \quad p_c = \eta_c P_H^{\text{use}}, \quad p_s = (1 - \eta_c) P_H^{\text{use}}, \quad (17)$$

$$p_{m,n} = \eta_n p_c, \quad p_{m,f} = (1 - \eta_n) p_c, \quad p_J = \eta_J P_J^{\max}, \quad (18)$$

$$f_H = f_H^{\min} + \eta_f (f_H^{\max} - f_H^{\min}). \quad (19)$$

This enforces HAPS power, NOMA ordering, AAV power, and CPU box constraints. Invalid RIS and mobility actions are masked according to b_R, b_J .

5 Physics-based action completion

The complete physical action is

$$\mathbf{a}_t = \mathcal{C}_{\text{phy}}(\mathbf{s}_t, \mathbf{a}_t^H; \hat{\mathcal{H}}_t), \quad (20)$$

where full estimated CSI $\hat{\mathcal{H}}_t$ remains in the numerical beam module and need not be serialised into the language prompt.

5.1 RIS and communication beam

Use a codebook

$$\mathcal{B}_R = \{\boldsymbol{\Theta}^{(1)}, \dots, \boldsymbol{\Theta}^{(C_R)}\} \quad (21)$$

formed from quantised geometric steering, DFT beams, or clustered offline alternating-optimisation solutions. The index produces the selected codeword or retains $\boldsymbol{\Theta}[t-1]$ when $c_R = 0$.

Define

$$\mathbf{R}_c = \zeta_T \hat{\mathbf{g}}_T \hat{\mathbf{g}}_T^H + \zeta_e \sum_{k \in \{n, f\}} \mathbf{C}_{m,k} + \mu_c \mathbf{I}. \quad (22)$$

A leakage-aware regularised matched beam is

$$\tilde{\mathbf{v}}_m = \mathbf{R}_c^{-1} \left(\alpha_n \frac{\hat{\mathbf{g}}_{m,n}}{\|\hat{\mathbf{g}}_{m,n}\|} + \alpha_f \frac{\hat{\mathbf{g}}_{m,f}}{\|\hat{\mathbf{g}}_{m,f}\|} \right), \quad \mathbf{v}_m = \frac{\tilde{\mathbf{v}}_m}{\|\tilde{\mathbf{v}}_m\|}, \quad (23)$$

$$\alpha_k = \frac{\exp(\beta_I \tilde{\Delta}_{m,k}^I)}{\sum_{j \in \{n, f\}} \exp(\beta_I \tilde{\Delta}_{m,j}^I)}. \quad (24)$$

This low-cost completion may later be replaced by a robust SCA/SDR subproblem.

5.2 Sensing and jamming beams

Using predicted target steering vectors,

$$\tilde{\mathbf{w}}_s = (\mathbf{C}_{\text{ang}} + \mu_s \mathbf{I})^{-1} \hat{\mathbf{a}}_t, \quad \mathbf{w}_s = \sqrt{p_s} \frac{\tilde{\mathbf{w}}_s}{\|\tilde{\mathbf{w}}_s\|}. \quad (25)$$

For the AAV, define

$$\mathbf{R}_J = \sum_{k \in \{n, f\}} \xi_k \hat{\mathbf{f}}_{J,m,k} \hat{\mathbf{f}}_{J,m,k}^H + \xi_H \mathbf{H}_{JH}^H \mathbf{H}_{JH} + \mu_J \mathbf{I}, \quad (26)$$

and set

$$\tilde{\mathbf{u}}_J = \mathbf{R}_J^{-1} \hat{\mathbf{f}}_{J,T}, \quad \mathbf{u}_J = \frac{\tilde{\mathbf{u}}_J}{\|\tilde{\mathbf{u}}_J\|}. \quad (27)$$

With

$$\mathbf{R}_I = \rho_{SI} \mathbf{H}_{SI} \mathbf{Q}_H \mathbf{H}_{SI}^H + p_J \mathbf{H}_{JH} \mathbf{u}_J \mathbf{u}_J^H \mathbf{H}_{JH}^H + \sigma_s^2 \mathbf{I}, \quad (28)$$

the sensing combiner is

$$\mathbf{u}_s = \frac{\mathbf{R}_I^{-1} \hat{\mathbf{a}}_r}{\|\mathbf{R}_I^{-1} \hat{\mathbf{a}}_r\|}. \quad (29)$$

Two or three alternating updates of (27) and (29) are sufficient for the deterministic module.

5.3 AAV command

$$\tilde{\mathbf{v}}_J[t] = \nu_J[t] [\cos \psi_J[t], \sin \psi_J[t], 0]^T. \quad (30)$$

It is subsequently projected onto speed, acceleration, flight-region, and safety constraints.

6 Feasibility projection and repair

Define

$$\mathbf{a}_t^{\text{safe}} = \mathcal{R}(\mathbf{s}_t, \mathbf{a}_t) \quad (31)$$

through this ordered procedure:

1. apply categorical masks and clip normalised continuous actions;
2. construct powers using equation (19);
3. project AAV velocity onto speed, acceleration, region, and separation sets;
4. evaluate far-user decoding and near-user SIC;
5. if SIC fails, reduce η_n by bisection until $\gamma_{m,n \rightarrow f} \geq \Gamma_{m,f}^{\text{SIC}}$;
6. if the pair remains infeasible, try the next teacher-ranked pair;
7. if sensing has a hard minimum, increase p_s by bounded line search;
8. if no payload action is feasible, use the fallback ($m_t = 0, p_c = 0, p_s = P_H^{\text{safe}}$).

For a convex admissible velocity set $\mathcal{V}_t$,

$$\mathbf{v}_J^{\text{safe}} = \arg \min_{\mathbf{v} \in \mathcal{V}_t} \|\mathbf{v} - \tilde{\mathbf{v}}_J\|_2^2. \quad (32)$$

For non-convex no-fly regions, select the closest feasible candidate from a small heading codebook. Log the original and repaired actions, with

$$d_{\text{rep}}[t] = \|\mathbf{a}_t^{\text{safe}} - \mathbf{a}_t\|_{\mathbf{W}}^2 \quad (33)$$

as a teacher-quality and deployment-safety diagnostic.

7 Reasoning teacher

7.1 Structured prompt

The teacher receives normalised state, objective, constraints, valid-action masks, and action semantics. Raw complex CSI is replaced by pair/global features; full CSI remains inside the verifier.

```

ROLE: Offline planning teacher for a causal HAPS-ISAC controller.
GOAL: Minimise H-step expected cost. Hard constraints have priority.

STATE:
pair_tokens: [...14 values...], ...]
global_token: [...24 values...]
virtual_queues: [...]
previous_ris_code: integer
valid_ris_codes: [...]
mobility_update_allowed: true/false

ACTION:
pair in {0,...,M}
ris_code in valid_ris_codes
eta_H, eta_c, eta_J, eta_f in [0,1]
eta_n in [0,0.5]
heading in [-1,1] (scaled by pi)
speed in [0,1] (scaled by Vmax)

RETURN:
K distinct candidate actions in strict JSON.
Include a short decision summary and no unrequested text.

```

The response schema is:

```

{
  "candidates": [
    {
      "pair": 2,
      "ris_code": 11,
      "continuous": [0.82,0.66,0.31,0.74,-0.20,0.55,0.70],
      "priority_tags": ["AoI","security"],
      "summary": "Serve pair 2; preserve sensing and raise jamming."
    }
  ]
}

```

The summary is optional auxiliary supervision. The verified action and rollout score, not the explanation, are the authoritative labels.

7.2 Candidate generation

For each state, generate

$$\tilde{\mathcal{A}}_t = \{\tilde{\mathbf{a}}_{t,1}^H, \dots, \tilde{\mathbf{a}}_{t,K}^H\}. \quad (34)$$

Use different seeds/temperatures or request age-first, sensing-first, security-first, and energy-first modes. Remove duplicate actions within tolerance ε_{dup} .

7.3 Stochastic rollout verification

Complete and repair every candidate, then evaluate it with N_{MC} common-random-number rollouts of horizon H . For rollout n ,

$$C_{i,n}^{(H)} = \sum_{h=0}^{H-1} \gamma^h \left[c(\mathbf{s}_{t+h}, \mathbf{a}_{t+h}) + \sum_{j=1}^{N_g} \lambda_j[t] g_j(\mathbf{s}_{t+h}, \mathbf{a}_{t+h}) \right] + \gamma^H V_\phi(\mathbf{s}_{t+H}). \quad (35)$$

c is the normalised stage cost corresponding to equation (1). After the first action, use a rollout policy π_{roll} equal to the current student, a conventional heuristic, or a mixture. Define

$$\hat{C}_i = \frac{1}{N_{\text{MC}}} \sum_{n=1}^{N_{\text{MC}}} C_{i,n}^{(H)}, \quad (36)$$

$$\hat{C}_i^{\text{risk}} = \hat{C}_i + \beta_{\text{risk}} \text{CVaR}_\alpha(\{C_{i,n}^{(H)}\}_{n=1}^{N_{\text{MC}}}). \quad (37)$$

Common random seeds reduce ranking variance. If V_i^{hard} is the weighted hard-violation magnitude, select lexicographically:

$$i^* = \arg \min_i (V_i^{\text{hard}}, \hat{C}_i^{\text{risk}}, d_{\text{rep},i}). \quad (38)$$

Therefore, a lower-cost infeasible candidate never defeats a feasible one.

7.4 Iterative feedback

For up to R_{fb} rounds, return concise evaluator feedback:

Candidate 2 is feasible and best.
Candidate 1: SIC margin -0.18; reduce near-user fraction.
Candidate 3: sensing outage 0.42; increase sensing fraction.
Best H-step cost: 0.684. Produce 4 improved candidates.

Stop when relative improvement is below ε_{fb} for two rounds. The method should be described as reasoning-assisted verified planning, not as proof that private chain-of-thought produces an optimum.

8 Verified demonstration dataset

Each record is

$$\mathcal{D}_i = (s_i, \mathcal{M}_i, \{\mathbf{a}_{i,k}^{\text{H}}, \hat{C}_{i,k}, V_{i,k}^{\text{hard}}, d_{\text{rep},i,k}\}_{k=1}^{K_i}, \mathbf{a}_i^*, G_i^*, \mathbf{y}_i^{\text{tag}}), \quad (39)$$

where $\mathcal{M}_i$ contains action masks, $G_i^* = -\hat{C}_{i^*}$, and $\mathbf{y}_i^{\text{tag}}$ is an optional multi-label priority tag. Store the simulator version, seeds, normalisation version, teacher identifier, prompt version, and repair-module version.

Preserve useful non-best candidates through

$$q_{i,k} = \frac{\mathbf{1}\{V_{i,k}^{\text{hard}} = V_{i,\min}^{\text{hard}}\} \exp[-(\hat{C}_{i,k} - \hat{C}_{i,\min})/\tau_q]}{\sum_{\ell} \mathbf{1}\{V_{i,\ell}^{\text{hard}} = V_{i,\min}^{\text{hard}}\} \exp[-(\hat{C}_{i,\ell} - \hat{C}_{i,\min})/\tau_q]}. \quad (40)$$

These weights retain multimodality when several decisions perform similarly.

9 Student policy

9.1 Recommended numerical architecture

Use a compact set/transformer policy:

1. shared pair encoder $\phi_{\text{p}} : \mathbb{R}^{14} \rightarrow \mathbb{R}^{64}$, implemented as $14 \rightarrow 64 \rightarrow 64$ with layer normalisation and GELU;
 2. global encoder $\phi_{\text{g}} : \mathbb{R}^{24+d_z+d_c} \rightarrow \mathbb{R}^{64}$;
 3. tokens $\mathbf{T} = [\phi_{\text{g}}; \phi_{\text{p}}(\mathbf{o}_1); \dots; \phi_{\text{p}}(\mathbf{o}_M)] \in \mathbb{R}^{(M+1) \times 64}$;
 4. two transformer blocks, four heads, width 64, feed-forward width 256;
 5. pooled representation $\mathbf{h}_t \in \mathbb{R}^{64}$;
 6. pair, RIS, continuous-action, value, constraint-risk, and optional priority-tag heads.
- Pair logits are computed from pair tokens, permitting variable M :

$$l_m^{\text{sch}} = \mathbf{w}_{\text{sch}}^{\text{T}}[\mathbf{h}_t; \mathbf{T}_m] + b_{\text{sch}}. \quad (41)$$

The RIS head produces $C_{\text{R}} + 1$ masked logits. The continuous head produces $\boldsymbol{\mu}_\theta, \log \boldsymbol{\sigma}_\theta \in \mathbb{R}^7$.

9.2 Continuous-action transform

Sample $\mathbf{u} \sim \mathcal{N}(\boldsymbol{\mu}_\theta, \text{diag}(\boldsymbol{\sigma}_\theta^2))$ and map

$$[\eta_{\text{H}}, \eta_{\text{c}}, \eta_{\text{J}}, \eta_{\text{f}}, \nu_{\text{J}}/V_{\text{J}}^{\text{max}}] = \text{sigmoid}([u_1, u_2, u_4, u_7, u_6]), \quad (42)$$

$$\eta_{\text{n}} = 0.5 \text{sigmoid}(u_3), \quad \psi_{\text{J}} = \pi \tanh(u_5). \quad (43)$$

At deterministic deployment, use $\mathbf{u} = \boldsymbol{\mu}_\theta$. A von Mises heading distribution may replace tanh if angular wrap-around is problematic.

9.3 Why not deploy a small language model?

For a fixed objective, the student consumes numerical tensors and emits a fixed schema. A numerical transformer is faster and easier to constrain. A small language model is justified only if natural-language intents or new constraint descriptions must be accepted at deployment. Then use a small intent encoder to produce $\mathbf{c}[t]$ while retaining numerical physical control.

10 Offline distillation

Let π_θ^{sch} and π_θ^{R} be categorical policies and π_θ^{cont} the transformed continuous density. Use

$$\mathcal{L}_\pi = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \sum_{k=1}^{K_i} q_{i,k} [\log \pi_\theta^{\text{sch}}(m_{i,k} | \mathbf{s}_i) + \log \pi_\theta^{\text{R}}(c_{i,k} | \mathbf{s}_i) + \log \pi_\theta^{\text{cont}}(\mathbf{a}_{i,k}^{\text{cont}} | \mathbf{s}_i)]. \quad (44)$$

Auxiliary losses are

$$\mathcal{L}_V = \frac{1}{|\mathcal{B}|} \sum_i \text{Huber}(V_\phi(\mathbf{s}_i) - G_i^*), \quad (45)$$

$$\mathcal{L}_{\text{risk}} = -\frac{1}{|\mathcal{B}|} \sum_i \sum_j [y_{i,j}^{\text{viol}} \log \hat{p}_{i,j} + (1 - y_{i,j}^{\text{viol}}) \log(1 - \hat{p}_{i,j})], \quad (46)$$

$$\mathcal{L}_{\text{tag}} = \text{BCE}(\hat{\mathbf{y}}_i^{\text{tag}}, \mathbf{y}_i^{\text{tag}}). \quad (47)$$

The complete objective is

$$\mathcal{L}_{\text{off}} = \mathcal{L}_\pi + \lambda_V \mathcal{L}_V + \lambda_{\text{risk}} \mathcal{L}_{\text{risk}} + \lambda_{\text{tag}} \mathcal{L}_{\text{tag}} + \lambda_2 \|\theta\|_2^2. \quad (48)$$

Set $\lambda_{\text{tag}} = 0$ if tags are unreliable. Do not reproduce verbose rationales unless explanation is an explicit evaluation target.

11 On-policy dataset aggregation

Offline demonstrations do not cover all student-visited states. During student rollouts, compute

$$u(\mathbf{s}) = \alpha_1 \mathcal{H}(\pi_\theta^{\text{sch}}) + \alpha_2 \mathcal{H}(\pi_\theta^{\text{R}}) + \alpha_3 \text{tr}(\Sigma_\theta^{\text{cont}}) + \alpha_4 \max_j \hat{p}_j^{\text{viol}} + \alpha_5 d_{\text{OOD}}(\mathbf{s}, \mathcal{D}). \quad (49)$$

Query the teacher only when $u(\mathbf{s}) > \tau_{\text{query}}$ or the fallback is invoked. Aggregate verified corrections:

$$\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}, \text{verified teacher correction})\}. \quad (50)$$

Repeat for N_{agg} rounds. This concentrates expensive calls on hard, deployment-relevant states.

12 Constrained reinforcement-learning refinement

After distillation, initialise a hybrid-action PPO policy. Let the environment reward be $r_t^{\text{env}} = -c_t$. The clipped surrogate is

$$\mathcal{L}_{\text{clip}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (51)$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t^{\text{H}} | \mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t^{\text{H}} | \mathbf{s}_t)}. \quad (52)$$

For hybrid actions, log probability is the sum of pair, RIS, and continuous log probabilities. Optimise

$$\max_{\theta} \quad \mathcal{L}_{\text{clip}} + \beta_{\mathcal{H}} \mathbb{E}[\mathcal{H}(\pi_\theta)] - \lambda_{\text{BC}}(n) \mathcal{L}_\pi - \sum_{j=1}^{N_g} \lambda_j \hat{\mathcal{J}}_{g_j}(\theta), \quad (53)$$

with

$$\lambda_j \leftarrow \left[\lambda_j + \eta_{\lambda} (\hat{\mathcal{J}}_{g_j} - d_j) \right]^+. \quad (54)$$

$\lambda_{\text{BC}}(n)$ decays but remains non-zero initially to prevent abrupt departure from verified behaviour. The safety layer remains active.

13 End-to-end training procedure

Phase A: simulator and deterministic modules

1. implement the causal environment and unit-test every SINR, age, tracking, delay, energy, and constraint recursion;
2. implement the RIS codebook, beam completion, repair, and fallback;
3. validate reduced cases using enumeration or conventional optimisation.

Phase B: initial teacher dataset

1. sample states from random, greedy, and conventional policies;
2. stratify by age, uncertainty, energy, and channel regime;
3. generate K candidates, complete/repair them, and run common-random rollouts;
4. perform up to R_{fb} feedback rounds and save full records.

Phase C: offline student

1. fit normalisers only on the training partition;
2. split by scenario seed/topology, not adjacent slots;
3. train with equation (48);
4. early-stop on held-out feasible return and violation rate.

Phase D: correction and RL

1. roll out the student with safety projection;
2. select hard states with equation (49);
3. query verified corrections and aggregate;
4. initialise constrained PPO and retain the best feasible validation checkpoint, not the largest unconstrained reward.

14 Deployment controller

At each slot:

1. update CSI, ages, target belief, energy, and virtual queues;
2. construct/normalise $\mathbf{s}_t$ and run one student forward pass;
3. apply masks and deterministic action decoding;
4. complete beams/RIS using $\mathcal{C}_{phy}$;
5. repair using $\mathcal{R}$ and execute fallback if a check fails;
6. apply the action and update the environment.

The large teacher is absent from real-time deployment.

15 Reference pseudocode

Verified teacher-data generation

```
for scenario in training_scenarios:
    env.reset(scenario)
    for t in range(T_collect):
        state_full = env.observe()
        state_num, prompt_state, masks = build_state(state_full)

        candidates = teacher.generate(
            prompt=serialize(prompt_state, masks),
            num_candidates=K
        )
        candidates = parse_deduplicate_and_mask(candidates)

        records = []
        common_seeds = sample_rollout_seeds(N_MC)
        for high_action in candidates:
            full_action = physics_complete(
                state_full, high_action, full_csi=env.csi
            )
            safe_action, repair_log = repair(
                state_full, full_action
            )
            score = rollout_evaluate(
```

```

        env.clone(), safe_action, horizon=H,
        seeds=common_seeds, rollout_policy=current_student
    )
    records.append(
        pack(high_action, safe_action, repair_log, score)
    )

    best = lexicographic_select(records)
    for r in range(R_fb):
        feedback = directional_feedback(records, best)
        improved = teacher.refine(prompt_state, feedback)
        records += verify(improved, common_seeds)
        best_new = lexicographic_select(records)
        if converged(best, best_new):
            break
        best = best_new

    dataset.write(state_num, masks, records, best, versions)
    env.step(collection_policy(state_full))

```

Student training and deployment

```

student = HybridSetTransformer(config)
student.fit(dataset, loss=L_off)

for aggregation_round in range(N_agg):
    states = rollout_student(student, safety_layer=True)
    hard_states = select_by_query_score(states, budget=B_query)
    corrections = verified_teacher_label(hard_states)
    dataset.extend(corrections)
    student.fit(dataset, loss=L_off)

student = constrained_ppo(
    env, init_policy=student, safety_layer=True,
    dual_update=True, behaviour_regularizer=True
)

def deployment_step(observation):
    state, masks = build_numeric_state(observation)
    high_action = student.deterministic_action(state, masks)
    full_action = physics_complete(observation, high_action)
    safe_action, log = repair(observation, full_action)
    if not final_invariant_check(safe_action):
        safe_action = sensing_only_fallback(observation)
    return safe_action

```

16 Implementation defaults

Component	Pilot value	Scaling experiment
NOMA pairs M	4	2, 4, 6, 8
RIS codebook C_R	32	16, 32, 64, 128
Teacher candidates K	8	1, 4, 8, 16
Feedback rounds R_{fb}	2	0, 1, 2, 3
Rollout horizon H	10 slots	1, 5, 10, 20
Monte Carlo rollouts N_{MC}	8	1, 4, 8, 16
Discount γ	0.99	0.95, 0.99, 0.995
Initial verified states	5,000	expand to 25,000–100,000
Student width	64	32, 64, 128
Transformer blocks	2	1, 2, 4
Teacher query rule	top 5% uncertainty	1%, 5%, 10%, 20%
PPO clip ϵ	0.2	0.1, 0.2, 0.3

The pilot first tests whether verified teacher actions beat strong heuristics. Large-scale generation is justified only after that gate. Easy states can be labelled by a conventional solver; teacher calls are reserved for ambiguous mixed-objective states.

17 Evaluation protocol

17.1 Baselines

Compare:

1. random feasible and greedy age-based policies;
2. alternating optimisation/SCA with receding horizon where tractable;
3. PPO, SAC/TD3 with an appropriate hybrid-action treatment;
4. hierarchical DRL;
5. one-shot teacher without verification;
6. verified teacher without distillation;
7. offline distilled student;
8. distilled student with dataset aggregation;
9. distilled student with aggregation and constrained PPO;
10. exhaustive search or dynamic programming on a reduced instance.

17.2 Metrics

Report

$\overline{\Delta^I}$, $\overline{\Delta^L}$, $\overline{\Delta^S}$, $\overline{\text{tr}(\mathbf{P}^{\text{av}})}$,
 secrecy outage, sensing outage, delivery reliability, hard-violation rate,
 HAPS/AAV energy, oracle gap, repair/fallback frequency,
 latency, parameter count, memory, teacher calls per accepted label.

Use confidence intervals over independent topology/channel seeds. Test unseen target speeds, UE layouts, CSI errors, RIS sizes, objective weights, and AAV energy levels.

17.3 Essential ablations

Remove one component at a time:

1. simulator verification;
2. multi-step horizon ($H = 1$);
3. virtual queues;
4. physics-based action completion;
5. safety repair;
6. candidate diversity ($K = 1$);
7. feedback ($R_{\text{fb}} = 0$);
8. quality-weighted multi-candidate distillation;
9. on-policy aggregation;
10. constrained PPO refinement;
11. terminal-value bootstrap;
12. priority tags.

These distinguish gains from reasoning, search, physical completion, verification, and ordinary RL.

18 Failure modes and safeguards

Failure	Required mitigation
Malformed output	Strict parser; retry once; otherwise discard
Infeasible action	Deterministic completion and ordered repair
Teacher exploits simulator bug	Unit tests, invariants, analytical cross-checks
Greedy behaviour	H -step rollout and terminal value
Constraint drift	Virtual queues/duals in state
Student distribution shift	Student-state aggregation and constrained RL
RIS action explosion	Codebook and update masks
Beam hallucination	Never request raw complex beams
Teacher cost explosion	Active querying, caching, solver labels for easy states
False optimality claim	Use best verified candidate unless an oracle proves optimality
Out-of-range state	OOD detector, fallback, and logging

19 Recommended research claim

A simulator-verified reasoning-distillation framework for causal, multi-timescale HAPS-ISAC control, in which a large reasoning model proposes high-level mixed actions, physics-based modules complete the physical-layer variables, stochastic rollouts verify decisions, and a constrained on-policy student provides real-time deployment.

The claim is not that chain-of-thought text replaces optimisation. It is that expensive general reasoning, physical verification, and low-cost constrained control are combined in a reproducible teacher-student system.

20 Module interfaces for implementation

Module	Input	Output
State builder	Environment state, queues, context	Pair/global tensors and masks
Prompt serializer	Normalised state and action schema	Deterministic structured prompt
Teacher client	Prompt, seed, generation settings	Parsed candidate list
Physics completion	Candidate and full CSI	Full beam/RIS/mobility action
Repair	Full action and constraints	Safe action, repair log
Rollout evaluator	State, safe action, seeds	Cost, violations, next-state statistics
Dataset writer	Candidate/evaluation metadata	Versioned demonstration record
Student policy	State tensors and masks	Hybrid action distributions and value
Query selector	Student uncertainty/risk/OOD	Teacher-query flag
Deployment shield	Student action and physical state	Executable safe action

Recommended tests include dimensional assertions; deterministic replay under fixed seeds; power-conservation and age-recursion invariants; finite-value checks after every beam solve; schema tests; boundary tests for action transforms; and regression tests comparing metrics before and after code changes.

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